



Time: 2hrs

FM: 100 |PM:35

All questions carry 1 mark.

Mathematics

1. If A, B and C are non-empty sets and $A \subset B$ then $A \cap B$ is
 - a. Empty set
 - b. B
 - c. A
 - d. C
2. Domain of $\sqrt{2-x}$ is
 - a. $(-\infty, 2]$
 - b. $(2, \infty)$
 - c. $(-\infty, \infty)$
 - d. $(-\infty, 2)$
3. If $\log_2(x-7) = 1$ then x is
 - a. 8
 - b. 9
 - c. 9
 - d. not known
4. The value of $\sin(\tan^{-1}x)$ is
 - a. x
 - b. $\frac{1}{1+x^2}$
 - c. $\frac{1}{\sqrt{1-x^2}}$
 - d. $\frac{x}{\sqrt{1+x^2}}$
5. If $\operatorname{cosec}^2 x = \operatorname{cosec}^2 \alpha$ then general value of x is
 - a. $n\pi \pm \alpha$
 - b. $n\pi + \alpha$
 - c. $2n\pi \pm \alpha$
 - d. $n\pi + (-1)^n \alpha$
6. The sum of the series $5+10+15+\dots\dots+100$ is
 - a. 1501
 - b. 1005
 - c. 1050
 - d. 895
7. If 3, x, 7 are in H.P then $x =$
 - a. $\frac{5}{21}$
 - b. $\frac{21}{5}$
 - c. $\frac{3}{15}$
 - d. $\frac{15}{3}$
8. If for suitable matrices A,B; $AB=A$ and $BA=B$, then A^2 equals
 - a. I
 - b. A
 - c. B
 - d. 0
9. The value of the determinant $\begin{vmatrix} 1 & 2 & 0 \\ 2 & 4 & 1 \\ 3 & 6 & 2 \end{vmatrix}$ is
 - a. 1
 - b. 3
 - c. 0
 - d. 4

10. The value of $(1 - \omega)(1 - \omega^2)$, where ω is the cube root of a complex number is

- a. 0 b. 1 c. 2 d. 3

11. $[2(\cos 15^\circ + i \sin 15^\circ)]^6 =$

- a. 64 b. 0 c. $64i$ d. $-64i$

12. If the roots of the equation $ax^2 + bx + c = 0$ are in the ratio 3:4 then

- a. $12b^2 = 49ac$ b. $12a^2 = 49bc$
c. $9b^2 = 16ac$ d. $16a^2 = 9bc$

13. If $f(x-1) = (x+2)$, then $f(x^2) =$

- a. $x^2 + 3$ b. $x^2 + x + 1$ c. $x - 3$ d. $x^2 + 5$

14. The length of the tangent to the circle $x^2 + y^2 = 4$ from the point (5, 2) is

- a. 4 b. 6 c. 5 d. 8

15. The area of the triangle with vertices (1, -1), (-1, 1), (-1, -1) is

- a. $\frac{1}{2}$ b. -2 c. 2 d. 0

16. The length of perpendicular from (0, 0) to the line $x + y + 1 = 0$ is

- a. $\frac{1}{\sqrt{2}}$ b. $\sqrt{2}$ c. $-\frac{1}{\sqrt{2}}$ d. $-\sqrt{2}$

17. $\lim_{x \rightarrow 0} \frac{\sqrt{1 - \cos 2x}}{\sqrt{2}x}$

- a. 1 b. -1 c. 0 d. doesn't exist

18. If $y = 5e^{3x} + \sin x$ then $\frac{dx}{dy}$ is

- a. $5e^{3x} + \cos x$ b. $15e^{3x} + \cos x$
c. $15e^{3x} - \cos x$ d. $15e^{3x} - \sin x$

19. If $x = t + \sin t$, $y = 1 - \cos t$ then $\frac{dy}{dx}$ is equal to

- a. $\tan t$ b. $\tan \frac{t}{2}$ c. $-\tan t$ d. $-\tan \frac{t}{2}$

20. The curve $f(x) = x^4 - 2x^3 + 5$ is concave upward for

- a. $x < 1$ b. $x > 1$ c. $x > 0$ d. $x > 1$ or $x < 0$

21. $\int \frac{\sin \sqrt{x}}{\sqrt{x}} dx$

- a. $-2\cos\sqrt{x} + c$
c. $\sin x + c$

- b. $2\cos\sqrt{x} + c$
d. $2\sin\sqrt{x} + c$

22. $\int_{-1}^1 \frac{x^3}{1+x^2} dx$

a. 0

b. 1

c. ∞

d. None

23. The area of the region bounded by the curve $y^2 = x$, the y-axis and abscissa $y = 2$ and $y = 4$ is

a. $\frac{3}{56}$

b. $\frac{56}{3}$

c. $\frac{64}{3}$

d. 56

24. The equation of the normal to the curve $y = \sin x$ at $(0,0)$ is

a. $x = 0$

b. $y = 0$

c. $x + y = 0$

d. $x - y = 0$

25. The system of equation $kx - 2y = 0$ & $x + 3y = 0$ has a unique solution. Then,

a. $k \neq -2/3$

b. $k = -2/3$

c. $k = 0$

d. $k = 2/3$

English:

26. Enzyme is a _____ word.

a. Two-syllable

b. Three-syllable

c. Four-syllable

d. Both a and b

27. Beautiful' has stress on _____ syllable.

a. First

b. Second

c. Third

d. None

28. Antonym of Vicious is

a. violent b. energetic b. passive d. tranquil

29. Pick the similar word of 'Obsolete'

a. Antiquate b. obsess c. noisy d. adorn

30. Insert the missing vowel in these words: defin_te, po_gnant, exist_nce.

a) i-i-e b) a-i-e c) i-e-a d) e-c-e

31. The word "go" has the same vowel sound as the word.....

(a) nor (b) four (c) sew (d) do

Code for Change

32. Whom did you give it?
(a) at (b) for (c) from (d) to
33. The complaint is not founded _____ facts.
a. Upon b. on c. at d. over
34. Many parents are blind _____ the faults of their children, but KP Oli is angry _____ this remark.
a. On, at b. to, with c. for, for d. with, to
35. The best indirect form of : "Call the first witness", said the judge.
a. The judge commanded to call the first witness.
b. The judge commanded not to call the first witness.
c. The judge said to them to call the first witness.
d. The judge commanded them to call the first witness.
36. Which of the following is a noun?
a. thicken b. impoverish c. mercantile d. mercy
37. The suffix -ism can be used as a/an ____
a. verb b. noun c. adjective d. adverb
38. She sang well that she _____ the competition.
a. can win b. shall win
c. might win d. should win
39. The fear of blood is called.
a) chirophobia b) ergophobia
c) monomania d) hematophobia
40. A pack of cigarettes _____ available.
a. is b. were c. has d. are
41. To be at sea
a. to enjoy
b. to go for sight seeing
c. to get perplexed
d. to quarrel with
42. Let's have fun, _____?
a. shall you b. shall we c. don't we d. will we
43. They ate as many pancakes as _____.
a. he b. they c. us d. she
44. Hater of alcoholic drink –
a. Iconoclast b. Teetotaler
c. NonDrunkard d. Narcissist

45. To hold one's tongue

- a. to stop someone while speaking
- b. disallow a person to speak
- c. to be silent
- d. to ignore

46. Antonym of altruism is:

- a. Unaltruism b. groupism
- c. individualism d. communism

47. Correct meaning of "magnum opus" is

- a) Precious b) a great work
- c) miraculous d) significant

48. The two countries live harmony each other.

- a. in, at b. in, with c. at, with d. between, with

49. The lawyer is.....LLB.

- a. a b. an c. the d. none

50. Are you my friend or enemy? has a tone.

- (a) rising and rising (b) rising and falling
- (c) rising (d) falling and falling

Physics

51. The force is related with time by $F = at + bt^2$ then dimensional formula of b is

- a. T^{-2} b. LT^{-1} c. LT^{-2} d. MLT^{-4}

52. A 10kg stone is suspended with a rope of breaking strength 30kg wt. The minimum time in which stone can be raised through a height of 10m starting from rest is

- a. 0.5 S b. 1 S c. $\sqrt{\frac{2}{3}}$ S d. 2 S

53. An ideal gas at pressure P is compressed suddenly so the density of gas become n times the initial value. The final pressure of gas will be

- a. $n^{\gamma}P$ b. $n^{\gamma}P$ c. $n^{\gamma-1}P$ d. $n^{1-\gamma}P$

54. The waves of intensities I and 4I interfere to produce maximum intensity I_0 . the intensity at a point where phase difference is 90°

- a. 5I b. 4I c. 3I d. I

55. An isochoric process is one which take place at

- a. constant internal energy b. constant entropy
- c. constant volume d. constant pressure

56. Two satellites are orbiting around the earth in a circular orbits of same radius. One of them is 5 times heavier than other. Then ratio of their time periods will be
 a. 25:1 b. 10:1 c. 5:1 d. 1:1
57. Cloudy nights are usually warmer, because clouds
 a. don't absorb heat b. don't radiate heat
 c. have low thermal conductivity d. high thermal conductivity
58. In stationary waves
 a. energy is uniformly distributed
 b. energy is minimum at nodes and maximum at antinodes
 c. energy is maximum at nodes and minimum at antinodes
 d. alternating maximum and minimum energy is produced at antinodes & nodes
59. Electric field intensity due to infinite sheet of charge having surface charge density σ is E . The electric field due to an infinite conducting sheet of same surface charge density is
 a. $E/2$ b. E c. $2E$ d. $4E$
60. Bulb of power 50w & 100w are connected in series then 50w bulb will
 a. glow brighter b. glow less brighter c. equal brightness d. not glow
61. Cathode rays are projected at right angle to magnetic field then their trajectory is
 a. ellipse b. circle c. parabola d. hyperbola
62. When exposed to sunlight, thin films of oil on water often exhibit colours due to the phenomenon
 a. interference b. diffraction c. dispersion d. polarization
63. A wire of resistance R is stretched to double of its length then resistance of wire will be
 a. $R/4$ b. $R/2$ c. $2R$ d. $4R$
64. The maximum energy of electrons released in a photo cell is independent of
 a. Frequency of incident radiation b. Intensity of incident radiation
 c. Nature of cathode d. All of above
65. A sample of 10g initially decay then mass after two mean lives is
 a. 2.5g b. 3.7g c. 6.3g d. 1.35g
66. A soap bubble is formed in soap solution has excess pressure P in it. The soap bubble of same radius is formed in air has excess pressure of
 a. P b. $2P$ c. $P/2$ d. $4P$
67. The magnifying power of telescope can be increased by
 a. increasing focal length b. using eye piece of higher power
 c. using eye piece of low power d. increasing distance of object

68. A plano convex lens of focal length 16cm is made by material of refractive index 1.5 then radius of curved surface is
 a. 8 cm b. 16 cm c. 24 cm d. 32 cm
69. Induced emf in a coil depends upon
 (a) Conductivity of coil (b) Amount of flux linked
 (c) Rate of change in flux (d) Resistance of coil
70. A proton and an electron have same de Broglie's wavelength then
 a. Both have same KE b. Proton has more KE than electron
 c. Electron has more KE than proton d. Both have same velocity
71. A ball of mass 'm' is dropped from a height 'h' and another ball of mass '3m' is dropped from a height of '9h' then time to reach the ground will be.
 (a) 1 : 1 (b) 1 : 3 (c) 3 : 1 (d) 1 : 9
72. A thief is running on straight road with 9m/s. A policeman chase him on a motorcycle with 10m/s from the instant when separation is 100m. The time in which policeman catch the thief is
 a. 5s b. 10s c. 90s d. 100s
73. A Carnot engine takes 3×10^6 calorie of heat from a reservoir at 627°C and gives it to sink at 27°C . The work done by engine is
 a. $4.2 \times 10^6 \text{ J}$ b. $8.4 \times 10^6 \text{ J}$ c. $16.8 \times 10^6 \text{ J}$ d. $32 \times 10^6 \text{ J}$
74. The ratio of specific charge of a proton to that of an alpha particle is
 a. 1:4 b. 1:2 c. 4:1 d. 2:1
75. Powder clings to the skin because of
 a. Compression b. Cohesion c. Adhesion d. Capillarity

Chemistry

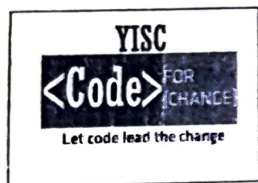
76. Equivalent weight of KMnO_4 is:
 a) 32.6 b) 52.6 c) 31.6 d) 49.0
77. Diagonal relationship exist between
 a) Li and Mg b) K and Mg
 c) Na and Mg d) Al and Mg
78. Order of dipole moment:
 a) $\text{HF} < \text{H}_2\text{S} < \text{H}_2\text{O}$ b) $\text{H}_2\text{S} < \text{HF} < \text{H}_2\text{O}$
 c) $\text{H}_2\text{O} < \text{HF} < \text{H}_2\text{S}$ d) $\text{HF} < \text{H}_2\text{O} < \text{H}_2\text{S}$
79. NH_3 has _____ geometry.
 a) Linear b) Trigonal Planar
 c) Octahedral d) Trigonal Pyramidal
80. The ratio of sigma and pi bonds in Benzene is
 a) 4 b) 6 c) 3 d) 5

81. What type of this reaction is?

$$\text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2 \rightarrow 6\text{CO}_2 + 6\text{H}_2\text{O} + \text{Energy}$$
 a) Oxidation reaction b) Reduction reaction
 c) Redox reaction d) Energetic reaction
82. The equivalent mass of $\text{K}_2\text{Cr}_2\text{O}_7$ when it acts as oxidizing agent in acidic medium is equal to
 a) M b) M/2 c) M/6 d) M/3
83. Which of the following decreases with the increase in temperature?
 a) Molality b) Mole Fraction
 c) Molarity d) Mole number
84. How many ml of 1M H_2SO_4 is required to neutralize 10 ml of 1M NaOH?
 a) 2 ml b) 5 ml c) 2.5 ml d) 10 ml
85. Which of the following is double salt?
 a) White Vitrol b) Epsom salt c) Mohr's salt d) None
86. The unit of equivalent conductivity is
 a. $\text{ohm}^{-1}\text{cm}^2 (\text{gm. eq})^{-1}$ b. $\text{ohm}^{-1}\text{cm}^2$
 c. ohm cm^2 d. $\text{S cm}^2\text{mol}^{-1}$
87. Aluminium metal is purified by
 a. Hoope's process b. Hall's process
 c. Serpeck's process d. Baeyer's process
88. Producer gas is the mixture of
 a. $\text{CO} + \text{N}_2$ b. CO and H_2
 c. $\text{N}_2 + \text{NH}_3$ d. CO , H_2 and N_2
89. Aqua – regia is a mixture of
 a. $3\text{HCl} + 1\text{HNO}_3$ b. $1\text{HCl} + 3\text{HNO}_3$
 c. $3\text{HNO}_3 + 1\text{H}_2\text{SO}_4$ d. $3\text{H}_2\text{SO}_4 + 1\text{HNO}_3$
90. Which of these is called ion exchanger for removing hardness?
 a. Calgon b. Permutit c. Organic d. Washing soda
91. The molten iron drawn from blast furnace is
 a. Wrought iron b. Pig iron c. Bessemer iron d. Stainless steel
92. Methane is formed from the hydrolysis of:
 a) Calcium carbide b) Methanol
 c) Aluminum carbide d) Ethanol
93. Reverse esterification is called:
 a. Hydrolysis b. trans-esterification
 c. Reduction d. Heating

94. Aniline reacts with acetaldehyde to form which:
 a) Schiff's base b) Carbylamine
 c) Imine d) Diazonium salt
95. Aspirin is:
 a) Antacid b) Analgesics c) Antipyretics d) both b & c
96. Mordant dye includes:
 a) Indigo b) Azo dye c) Alizarin d) Martius yellow
97. Chalcopyrite is an ore of
 a) Fe b) Cu c) Zn d) Hg
98. The unit of rate constant for first order reaction is
 (a) mol s^{-1} (b) $\text{L mol}^{-1} \text{s}^{-1}$ (c) s^{-1} (d) $\text{mol L}^{-1} \text{s}^{-1}$
99. The p^H of 0.01 M NaOH solution is
 a) 2 b) 0.1 c) 12 d) 0.01
100. KO_2 is an example of
 a) normal oxide b) sub oxide c) superoxide d) peroxide

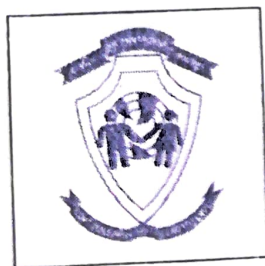
-Best of Luck-



Organized by:

"Code for Change" is a project initiated by "Youth in Social Change" in collaboration with the students from various colleges around the country with an aim to bring the IT students and the industry professionals together. We believe that, students should be provided with **relevant skills** to prepare them for their career. Realizing this fact, we have initiated "Code for Change" to provide platform and opportunities for them. In the past two years we have conducted nationwide hackathons, workshops and training programs targeting the IT students.

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